IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity

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Activity 18: Texture Analysis

Match the feature to its meaning:

Contrast → Measures the difference between dark and bright pixels in a texture.

Entropy → Measures the randomness or complexity of the texture.

Energy → Measures the uniformity or regularity of the texture.

Homogeneity → Measures how similar neighboring pixel values are.

Which texture type has high entropy? Why?

A highly irregular or complex texture has high entropy because it contains a large variety of pixel intensities and greater randomness. There is less uniformity in the texture, making the distribution of pixel values more unpredictable. Examples include rough surfaces, natural scenes, and highly detailed patterns.